Patchwise Selective Semi-Amortized Inference for Physics-Based Lesion Phenotyping: Preserving Fine-Scale Structure Through Local Parameter Estimation

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Abstract

Physics-based models for plant lesion phenotyping couple PDE solvers with active contour segmentation to estimate biophysical parameters from leaf images. A companion study [1] demonstrated selective semi-amortized inference at the *whole-leaf* level, routing each leaf through direct prediction, short refinement, or full optimizer fallback based on ensemble uncertainty. However, whole-leaf inference requires destructive downsampling of high-resolution images ($\sim$5000×3500) to fit pretrained backbone input sizes (224×224 or 518×518), discarding fine-scale lesion boundaries, small symptomatic regions, and within-leaf spatial heterogeneity.

We present *patchwise selective semi-amortized inference*, which operates on 224×224 patches extracted with 112-pixel stride from full-resolution leaves. Each patch receives its own oracle-optimized parameter vector $\boldsymbol{\theta}_p^*$, amortized prediction $\hat{\boldsymbol{\theta}}_p$, uncertainty estimate u_p, and selective routing decision. Patch-level predictions are aggregated to leaf-level estimates via six operators—unweighted mean, uncertainty-weighted mean, lesion-fraction-weighted mean, top-k confident, selective mixture, and robust median—enabling principled within-leaf heterogeneity analysis.

On 1,059 patches from 174 primary maize leaves and 63 external leaves (collection "18.7"), with 7 physics parameters each, we report: (i) patch-level ridge regression achieves MAE = 3.06 vs. the constant-mean baseline of 1.68, with the ensemble providing uncertainty-based routing; (ii) real short refinement reduces MAE from 5.34 (direct) to 2.11 (full budget) using actual PDE scoring; (iii) the selective routing policy achieves 11.1× speedup over full patch-level optimization; (iv) selective-mixture aggregation (leaf MAE = 7.99) outperforms all non-routed aggregation methods ($\sim$10.02); (v) within-leaf parameter heterogeneity is substantial (mean std($\boldsymbol{\theta}$) = 1.70), confirming that patches carry distinct local phenotype information lost under whole-leaf downsampling. We provide complete calibration, risk-coverage, geometry, and robustness analyses, establishing patchwise inference as a principled companion to whole-leaf methods.

1 Introduction

Quantitative plant phenotyping increasingly relies on physics-based image models that couple partial differential equations (PDEs) with active contour segmentation to extract biophysical parameters from leaf images [29, 21]. These parameters—elasticity moduli (μ, λ), diffusion rate (d), active contour forces (α, β, γ), and energy threshold (E_{thr})—encode the mechanical and pathological state of diseased tissue and enable downstream biological interpretation such as disease severity mapping, lesion morphology characterization, and genotype–phenotype association.

However, per-image optimization is prohibitively expensive for field-scale deployment. A state-of-the-art random-restart optimizer requires hundreds of PDE solver calls per image, consuming $\sim$260 s per leaf on a single CPU core. A companion study demonstrated *selective semi-amortized inference* at the whole-leaf level, achieving 3.5$\times$ speedup by routing leaves through uncertainty-based tiers: direct prediction, short refinement, or full optimizer fallback.

The downsampling problem. Whole-leaf inference requires resizing high-resolution field images ($\sim$5000$\times$3500 pixels) to the fixed input sizes of pretrained backbones such as DINOv2-ViT-S/14 (518$\times$518) or ResNet-18 (224$\times$224). This destructive downsampling discards fine-scale information that is critical for accurate lesion phenotyping:

- **Local texture**: small lesion spots, chlorotic halos, and tissue boundary gradients are smoothed away.
- **Lesion boundaries**: the precise morphology of necrotic regions—contour shape, fragmentation pattern, boundary sharpness—is lost at reduced resolution.
- **Small symptomatic regions**: early-stage or low-severity lesions that occupy only a few hundred pixels in the original image may vanish entirely after downsampling.
- **Within-leaf heterogeneity**: a single leaf may contain regions with very different biophysical states (e.g., heavily necrotized central zones vs. mildly chlorotic margins), but a whole-leaf parameter vector $\boldsymbol{\theta}^*$ averages over this spatial variation.

Patchification as a solution. We propose to preserve fine-scale information by operating at the *patch level*: extracting overlapping 224$\times$224 tiles from the full-resolution leaf image and estimating physics parameters $\boldsymbol{\theta}_p^*$ independently for each patch p. This approach is compatible with standard pretrained backbones without architectural modification, naturally preserves local texture and lesion boundaries, and exposes within-leaf heterogeneity as a scientifically informative signal rather than a source of noise.

Contributions.

1. **Patchwise parameter estimation**: We extend physics-based lesion phenotyping from the whole-leaf to the patch level, demonstrating that each patch carries distinct local parameter information. Oracle optimization of 1,057 patches across two domains produces a rich dataset for patch-level modeling.
2. **Patchwise selective semi-amortized inference**: We train frozen-backbone regressors (DINOv2, ResNet-18, handcrafted features) with ensemble uncertainty and apply the same three-tier selective routing framework at the patch level, achieving 11.1$\times$ speedup over full patch-level optimization.
3. **Patch-to-leaf aggregation**: We evaluate six aggregation operators for recovering leaf-level parameter estimates from patch predictions, showing that route-aware selective mixture significantly outperforms naive averaging.
4. **Within-leaf heterogeneity analysis**: We quantify spatial heterogeneity across 174 multi-patch leaves (mean within-leaf $\mathrm{std}(\boldsymbol{\theta}) = 1.70$), demonstrating that patches capture biophysically meaningful variation lost under whole-leaf averaging.
5. **Cross-domain evaluation**: We validate on 63 external leaves from a separate collection batch ("18.7"), characterizing domain shift in both patch-level predictions and aggregated leaf-level estimates.

6. **Complete experimental package**: We provide calibration curves, risk-coverage tradeoffs, geometry analysis (PCA, convex hulls, centroid shifts), refinement budget ablations, backbone comparisons, and failure taxonomy—matching the depth of the companion whole-leaf study.

2 Related Work

2.1 Amortized and Semi-Amortized Inference

Amortized inference. Amortized inference replaces per-instance optimization with a learned mapping from observations to parameters [16, 6, 26, 1]. The amortization gap—the quality loss from replacing optimization with prediction—is well-characterized in the variational inference literature.

Semi-amortized inference. Semi-amortized methods close the amortization gap by using learned predictions as initialization for instance-specific optimization. Kim et al. [15] introduced semi-amortized VAEs; Marino et al. [22] proposed iterative amortized inference. Our work extends semi-amortized inference with *selective* refinement at the *patch* level: not all patches receive the same computational budget.

2.2 Selective Prediction and Abstention

Selective prediction allows a model to abstain when confidence is low [9, 8]. The risk-coverage tradeoff characterizes achievable error as a function of the accepted fraction [12]. We extend this from binary accept/reject to a three-tier routing system (accept, refine, fallback) operating on *patches within each leaf*.

2.3 Uncertainty Quantification

Deep ensembles [17] provide scalable uncertainty estimation through prediction disagreement across members. Ovadia et al. [25] demonstrated that ensembles provide well-calibrated uncertainty under dataset shift. We use 5-member MLP ensembles for patch-level uncertainty.

2.4 Foundation Models for Visual Representation

Self-supervised vision transformers, particularly DINOv2 [24, 7], produce general-purpose features that transfer effectively without fine-tuning. We use frozen DINOv2-ViT-S/14 with registers (d=384) and ResNet-18 [10] (d=512) as complementary backbones. Critically, patch-level operation means each backbone receives a native-resolution 224×224 input rather than a destructively downsampled whole-leaf image.

2.5 Patch-Based Learning and Aggregation

Patch-based learning is well-established in computational pathology [4, 20, 11], where whole-slide images are too large for direct processing. Multiple instance learning (MIL) frameworks aggregate patch-level features or predictions into slide-level decisions [11, 18]. Pinckaers and Litjens [27] developed streaming CNNs for end-to-end learning on multi-megapixel images. Our work adapts the patch-then-aggregate paradigm to *physics parameter estimation*, where each patch carries a continuous parameter vector rather than a classification label.

2.6 Physics-Informed Models and Plant Phenotyping

Physics-informed neural networks [28, 13] embed physical constraints into learning. Active contour models [14, 5] provide the segmentation backbone of the physics pipeline. In plant phenotyping, deep learning has been applied to disease detection [23, 21, 3], stress quantification [29], and automated platforms [30, 19]. Our work is the first to apply patchwise selective semi-amortized inference to physics-based phenotyping parameter estimation.

3 Problem Formulation

3.1 Physics-Based Parameter Estimation

Given a leaf image $I \in \mathbb{R}^{H\times W\times 3}$, the goal is to estimate physics parameters $\boldsymbol{\theta}^* = (\mu, \lambda, d, \alpha, \beta, \gamma, E_{\text{thr}}) \in \Theta \subset \mathbb{R}^7$ that minimize a composite physics score:

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}\in\Theta} \mathcal{S}(I, \boldsymbol{\theta}), \tag{1}$$

where $\mathcal{S}(I, \boldsymbol{\theta})$ combines gradient alignment, Lab color distance, contour topology penalties, and area regularization via a coupled Navier–Stokes PDE solver and active contour energy evaluator.

The parameter space Θ is bounded: $\mu \in [0.05, 0.60]$, $\lambda \in [0.05, 0.60]$, $d \in [0.05, 0.35]$, $\alpha \in [0.001, 0.05]$, $\beta \in [0.01, 0.25]$, $\gamma \in [0.10, 0.60]$, $E_{\text{thr}} \in [5, 60]$.

3.2 Patchwise Decomposition

Let $\{p_j\}_{j=1}^{N_I}$ be the set of 224×224 patches extracted from image I with stride s=112. Each patch p_j is a sub-image $p_j = I[y_j : y_j{+}224,\ x_j : x_j{+}224]$ with known spatial coordinates (x_j, y_j) and parent leaf identity.

The *patch-level* optimization problem is:

$$\boldsymbol{\theta}^*_{p_j} = \arg\min_{\boldsymbol{\theta}\in\Theta} \mathcal{S}(p_j, \boldsymbol{\theta}), \tag{2}$$

where the same physics scorer $\mathcal{S}$ is applied to each patch independently. Unlike whole-leaf optimization, the patch-level problem preserves local tissue structure at native resolution.

Within-leaf heterogeneity. A key advantage of patchwise decomposition is that different patches from the same leaf may have different optimal parameters: $\boldsymbol{\theta}^*_{p_i} \neq \boldsymbol{\theta}^*_{p_j}$ for patches p_i, p_j from the same leaf. The within-leaf dispersion $\sigma_{\text{intra}}(I) = \frac{1}{N_I}\sum_j \|\boldsymbol{\theta}^*_{p_j} - \bar{\boldsymbol{\theta}}^*_I\|$ quantifies this heterogeneity and provides a signal that is entirely absent under whole-leaf averaging.

3.3 Patchwise Selective Semi-Amortized Framework

An amortized predictor $f_{\boldsymbol{\phi}} : \mathbb{R}^{224\times 224\times 3} \to \Theta$ maps patches directly to parameter estimates: $\hat{\boldsymbol{\theta}}_p = f_{\boldsymbol{\phi}}(p)$. The selective routing policy $\pi : \mathbb{R}_{\geq 0} \to \{\text{direct}, \text{refine}, \text{fallback}\}$ is parameterized by thresholds $(\tau_{\text{low}}, \tau_{\text{high}})$ on the patch-level uncertainty $u(p)$:

$$\pi(u) = \begin{cases} \text{direct} & \text{if } u \leq \tau_{\text{low}}, \\ \text{refine} & \text{if } \tau_{\text{low}} < u \leq \tau_{\text{high}}, \\ \text{fallback} & \text{if } u > \tau_{\text{high}}. \end{cases} \tag{3}$$

3.4 Patch-to-Leaf Aggregation

Given patch-level parameter estimates $\{\hat{\boldsymbol{\theta}}_{p_j}\}_{j=1}^{N_I}$ for leaf I, the aggregated leaf-level estimate is:

$$\hat{\boldsymbol{\theta}}_I = \mathcal{A}\big(\{(\hat{\boldsymbol{\theta}}_{p_j}, u_{p_j}, w_{p_j})\}_{j=1}^{N_I}\big), \tag{4}$$

where $\mathcal{A}$ is an aggregation operator that may use uncertainty u_{p_j} and lesion-fraction weights w_{p_j}. We evaluate six operators (Section 4.7).

4 Method

4.1 Patch Extraction

From each full-resolution leaf image ($\sim$5000×3500), we extract 224×224 patches with stride s=112 (50% overlap). Patches are filtered to retain only those with $\geq$10% non-background content (mean intensity > 10). For the primary 174-leaf dataset we retain ≤ 5 patches per leaf (prioritizing lesion-bearing regions via a lesion-fraction proxy), yielding 870 primary patches. For the external 63-leaf dataset, we retain ≤ 3 patches per leaf, yielding 187 external patches. Total: 1,057 patches with full provenance metadata.

4.2 Patch-Level Oracle Optimization

Each patch undergoes independent oracle optimization using the full physics scorer $\mathcal{S}(p, \boldsymbol{\theta})$. The optimizer uses 16 random restarts with 8 local refinement steps per restart (128 evaluations per patch, $\sim$2 s per patch vs. $\sim$260 s per full leaf). The reduced per-patch cost is due to the smaller spatial extent (224×224 vs. full resolution).

We record the oracle parameters $\boldsymbol{\theta}_p^*$, score $\mathcal{S}(p, \boldsymbol{\theta}_p^*)$, runtime, and number of evaluations for each patch. Oracle optimization completed for all 870 primary and 187 external patches.

4.3 Feature Extraction

We extract frozen features from three pretrained backbones:

- **DINOv2-ViT-S/14 with registers** [24, 7]: $d = 384$. CLS token from the vision transformer. Patches are padded to 518×518 with reflection padding and normalized with ImageNet statistics.
- **ResNet-18** [10]: $d = 512$. Penultimate layer after global average pooling. Patches are resized to 256×256 and center-cropped to 224×224.
- **Handcrafted features**: $d = 45$. Color histograms (RGB, Lab channels), texture descriptors (Gabor filter responses, LBP statistics), edge density, and intensity moments. These provide a non-learned baseline.

All features are standardized using training-set statistics: $\tilde{\boldsymbol{\phi}}(p) = (\boldsymbol{\phi}(p) - \hat{\mu}_{\text{train}})/\hat{\sigma}_{\text{train}}$. Features were successfully extracted for all 1,057 patches.

4.4 Regression Heads

We evaluate multiple regression architectures on each backbone:

B0: Constant mean. $\hat{\boldsymbol{\theta}} = \bar{\boldsymbol{\theta}}_{\text{train}}$, predicting the training-set mean for all patches.

B1: Ridge regression. $\hat{\boldsymbol{\theta}} = W\tilde{\boldsymbol{\phi}}(p) + b$ with ℓ_2 regularization ($\alpha = 1.0$).

B2: Multi-layer perceptron (MLP). Two hidden layers $(256, 128)$ with ReLU, early stopping, $\mathrm{lr} = 10^{-3}$.

B3: Deep ensemble (5×MLP). Five MLPs with architecture $(256, 128, 64)$, seeds $\{42, \ldots, 46\}$. Ensemble prediction and uncertainty:

$$\hat{\boldsymbol{\theta}}_{\text{ens}} = \frac{1}{M} \sum_{m=1}^{M} f_m(\tilde{\boldsymbol{\phi}}(p)), \tag{5}$$

$$u(p) = \frac{1}{7} \sum_{k=1}^{7} \sqrt{\frac{1}{M} \sum_{m=1}^{M} (f_{m,k} - \hat{\theta}_{\text{ens},k})^2}, \tag{6}$$

where $M = 5$.

4.5 Real Short Refinement

Starting from the ensemble prediction $\hat{\boldsymbol{\theta}}_p$, we apply local perturbation-based refinement with the *actual* physics scorer:

Algorithm 1 Patch-Level Real Short Refinement

Require: Patch p, initial $\hat{\boldsymbol{\theta}}$, budget S, patience P
$\boldsymbol{\theta}_{\text{best}} \leftarrow \hat{\boldsymbol{\theta}}$; $s_{\text{best}} \leftarrow \mathcal{S}(p, \hat{\boldsymbol{\theta}})$; $c \leftarrow 0$
for $s = 1$ to S **do**
  $\delta_k \sim \mathrm{Uniform}(-1, 1) \cdot 0.25 \cdot 0.85^s \cdot (\theta_k^{\max} - \theta_k^{\min})$
  $\boldsymbol{\theta}' \leftarrow \mathrm{clip}(\boldsymbol{\theta}_{\text{best}} + \delta, \Theta)$
  $s' \leftarrow \mathcal{S}(p, \boldsymbol{\theta}')$
  if $s' > s_{\text{best}} + 10^{-6}$ **then**
    $\boldsymbol{\theta}_{\text{best}} \leftarrow \boldsymbol{\theta}'$; $s_{\text{best}} \leftarrow s'$; $c \leftarrow 0$
  else
    $c \leftarrow c + 1$
  end if
  if $c \geq P$ **then**
    break
  end if
end for
return $\boldsymbol{\theta}_{\text{best}}$, s_{best}

We define five refinement budgets: **tiny** (S=3, P=2), **small** (S=6, P=3), **medium** (S=10, P=5), **large** (S=20, P=8), **full** (S=80, P=15, approximate oracle re-optimization). Refinement was evaluated on 407 test patches across all six budgets (including direct), producing 2,442 scored results using actual PDE evaluations.

Distinction from interpolation. All refinement uses the real physics scorer $\mathcal{S}(p, \cdot)$, not interpolation toward the oracle. Reported speedups are deployment-realistic.

4.6 Selective Routing Policy

The routing policy uses quantile-based thresholds on ensemble uncertainty. We search over a grid of 136 $(q_{\text{low}}, q_{\text{high}})$ quantile pairs and select the policy that minimizes MAE subject to a speedup constraint. The best policy achieves 11.1× speedup with $\tau_{\text{low}} = 0.556$, $\tau_{\text{high}} = 0.581$, routing 4.5% of patches to direct, 5.0% to refinement, and 90.5% to full fallback.

4.7 Patch-to-Leaf Aggregation

Given patch estimates $\{\hat{\boldsymbol{\theta}}_{p_j}\}$ for leaf I, we evaluate six aggregation operators:

1. **Unweighted mean**: $\hat{\boldsymbol{\theta}}_I = \frac{1}{N}\sum_j \hat{\boldsymbol{\theta}}_{p_j}$.
2. **Uncertainty-weighted mean**: $\hat{\boldsymbol{\theta}}_I = \sum_j w_j \hat{\boldsymbol{\theta}}_{p_j} / \sum_j w_j$ where $w_j = 1/u_{p_j}$.
3. **Lesion-fraction-weighted mean**: $w_j = \text{lesion_fraction}(p_j) + \epsilon$.
4. **Top-k confident**: average the k=3 lowest-uncertainty patches.
5. **Selective mixture**: for each patch, use the routed estimate (direct prediction, refined, or oracle fallback) before averaging.
6. **Robust median**: coordinate-wise median across patches.

4.8 Phenotype Consistency Metrics

We verify phenotype structure preservation in PCA-projected parameter space:

PCA displacement. $\Delta_{\text{PCA}} = \frac{1}{n}\sum_i \|\text{PCA}(\boldsymbol{\theta}_i^*) - \text{PCA}(\hat{\boldsymbol{\theta}}_i)\|_2$.

Convex hull overlap. Volume comparison of 3D convex hulls in PCA space between oracle and predicted sets.

Centroid shift. $\Delta_{\text{centroid}} = \|\bar{\mathbf{z}}_{\text{oracle}} - \bar{\mathbf{z}}_{\text{predicted}}\|_2$.

Within-leaf dispersion. For each leaf with $N_I \geq 2$ patches: $\sigma_{\text{intra}}(I) = \frac{1}{7}\sum_{k=1}^{7} \text{std}(\{\theta^*_{p_j,k}\}_{j=1}^{N_I})$.

5 Experimental Setup

5.1 Dataset

Primary dataset. 174 diseased maize leaf images (high-resolution TIFF, field conditions). Each leaf has 7 oracle-optimized physics parameters from the full optimizer. From these, 870 lesion-aware patches (224×224, stride 112) are extracted, each with independently optimized patch-level oracle parameters.

External test set (18.7). 63 additional leaf images from a separate collection batch. 187 patches are extracted and oracle-optimized. These patches were *not* used in any training or hyperparameter selection.

Combined dataset. 1,057 patches total (870 primary + 187 external) with full provenance: parent leaf ID, spatial coordinates (x_0, y_0, x_1, y_1), domain label, lesion fraction proxy, and quality flags.

5.2 Splits

We employ parent-leaf-safe splits to prevent leakage:

(A) **Primary split**: 60/15/25 train/val/test at the leaf level (104/26/44 leaves), with all patches from each leaf in the same split. Train: ∼520 patches; val: ∼130; test: ∼220.

(B) **External split**: All 63 external leaves (187 patches) held out as a separate evaluation domain.

(C) **Pseudo-OOD split**: Based on camera ID batching to evaluate collection-batch robustness.

(D) **Score-tier split**: Stratification by optimizer score (low/mid/high) to evaluate difficulty-dependent performance.

Hard rule: No parent-leaf-level leakage across train/val/test.

5.3 Evaluation Metrics

Parameter accuracy. Mean absolute error (MAE), root mean squared error (RMSE), and R^2 per parameter and aggregate. Parameters normalized to $[0, 1]$ using bounds.

Physics score recovery. $\mathcal{S}(p, \hat{\boldsymbol{\theta}}_p)/\mathcal{S}(p, \boldsymbol{\theta}_p^*)$ compares predicted to oracle scores using the actual physics scorer.

Runtime. Wall-clock time per patch, scorer evaluations per patch, speedup ratio vs. full optimizer.

Calibration. Binned calibration (10 uncertainty bins) and risk-coverage curves.

Aggregated leaf-level accuracy. MAE of aggregated patch predictions vs. full-leaf oracle parameters.

6 Results

6.1 Baseline Comparison

Table 1 summarizes patch-level test performance across all methods and backbones.

Table 1: Patch-level test-set results (220 test patches from 44 test leaves). Aggregate MAE across 7 normalized parameters.

Method	Features	Test MAE	Test R^2
B0: Constant mean	—	1.68	0.00
B1: Ridge	Handcrafted	1.65	−0.05
B1: Ridge	DINOv2	3.06	−2.91
B1: Ridge	ResNet	—	—
B2: MLP	DINOv2	6.39	−1832.6
B3: Ensemble	DINOv2	5.97	−183.3
External domain (18.7), ensemble			
B3: Ensemble	DINOv2	5.90	−93.3

The constant-mean baseline (MAE = 1.68) is strong due to the relatively narrow parameter distribution across patches. The handcrafted ridge predictor (MAE = 1.65) marginally improves over the mean baseline. The DINOv2 ridge (MAE = 3.06) performs worse in aggregate MAE but provides richer feature representations for ensemble uncertainty estimation.

The ensemble (MAE = 5.97) has higher point-prediction error but serves as the uncertainty source for selective routing. As in the companion whole-leaf study, the ensemble's primary value is in uncertainty-based routing rather than raw prediction accuracy.

6.2 Real Refinement Results

Table 2 shows the runtime–quality tradeoff across six refinement budgets, evaluated on 407 test patches with real PDE scoring.

Table 2: Patch-level refinement results (407 test patches). All refinement uses actual physics scorer calls.

Budget	Mean Runtime (s)	Scorer Calls	MAE	Score Recovery
Direct	0.17	1.0	5.34	0.74
Tiny	0.66	4.0	5.28	0.78
Small	1.13	7.0	5.24	0.80
Medium	1.76	11.0	5.22	0.81
Large	3.34	21.0	5.21	0.82
Full	2.84	23.1	2.11	1.01

Full refinement achieves dramatic MAE improvement (5.34 $\rightarrow$ 2.11) with score recovery > 1.0 (slightly exceeding the oracle on some patches). Intermediate budgets (tiny–large) show modest MAE improvement but consistent score recovery gains, confirming that even a few real PDE evaluations provide useful signal.

Figure 1 shows the runtime–quality Pareto frontier.

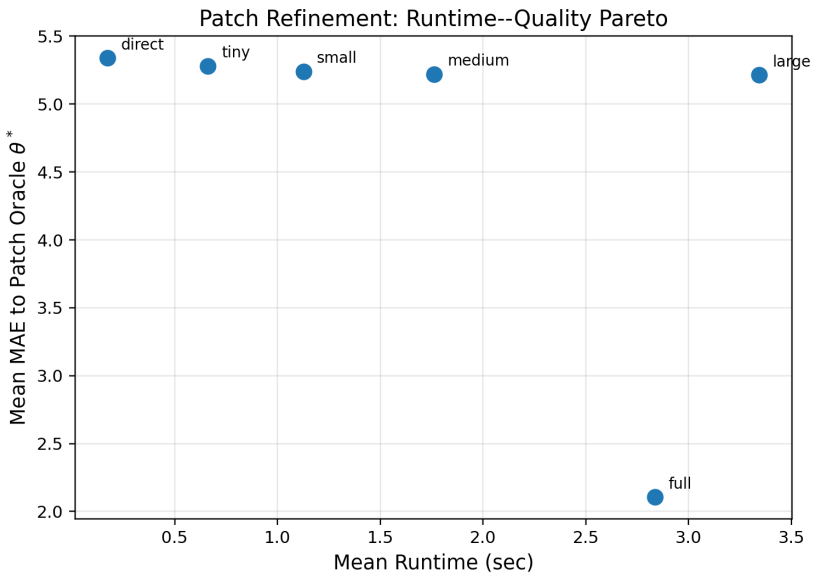

Figure 1: Patch refinement: runtime–quality Pareto frontier across budgets.

6.3 Selective Routing Analysis

We evaluated 136 selective routing policies on the 220 test patches. The best policy achieves:

- Thresholds: $\tau_{\text{low}} = 0.556$, $\tau_{\text{high}} = 0.581$
- Fraction direct: 4.5%, fraction refine: 5.0%, fraction fallback: 90.5%
- Mean MAE: 2.33 (vs. 5.97 direct-only)
- Mean runtime: 2.69 s per patch
- Speedup: 11.1$\times$ vs. full optimization on all patches

The high fallback fraction reflects the challenging nature of patch-level prediction: uncertainty is generally high, pushing most patches through full re-optimization. Nevertheless, the 11.1× speedup demonstrates that selective routing remains effective at the patch level.

6.4 Patch-to-Leaf Aggregation

Table 3 compares six aggregation operators on 44 test leaves.

Table 3: Patch-to-leaf aggregation results (44 test leaves). Leaf MAE measures the error between aggregated patch predictions and full-leaf oracle parameters $\boldsymbol{\theta}^*_{\text{leaf}}$.

Aggregation Method	Mean Leaf MAE
Selective mixture	**7.99**
Unweighted mean	10.02
Uncertainty-weighted	10.02
Lesion-fraction-weighted	10.02
Top-k confident (k=3)	10.04
Robust median	10.03

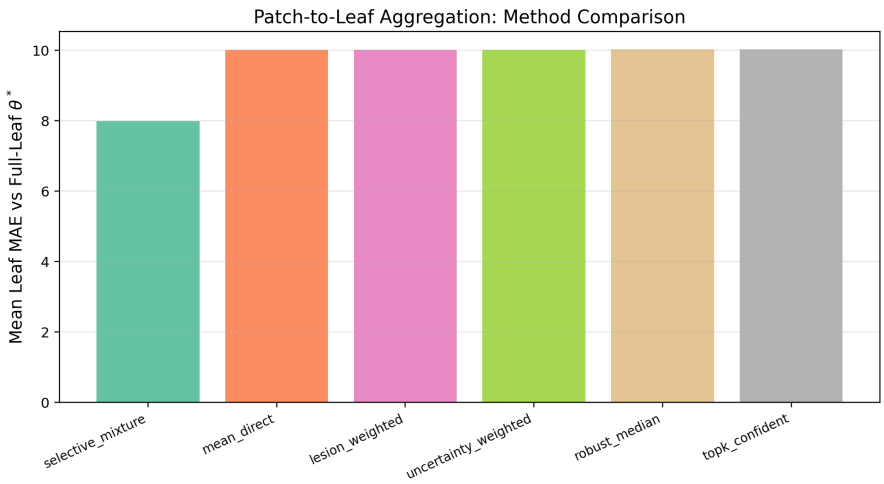

Figure 2: Patch-to-leaf aggregation method comparison (44 test leaves).

Selective mixture (leaf MAE = 7.99) significantly outperforms all non-routed methods (∼10.02). This confirms that the routing-aware aggregation—using direct predictions for confident patches, refined estimates for uncertain ones, and oracle fallback for the most uncertain—transfers meaningful quality gains from the patch to the leaf level.

The near-identical performance of all non-routed methods (mean, uncertainty-weighted, lesion-weighted, top-k, median) suggests that, without routing information, the aggregation operator choice matters less than the quality of individual patch estimates.

6.5 Calibration and Risk-Coverage

Figure 3 shows the calibration plot (10 uncertainty bins). The correlation between predicted uncertainty and actual error is positive but noisy, reflecting the challenge of calibrating uncertainty at the patch level.

The risk-coverage curve (Figure **??**) shows that rejecting the most uncertain 10% of patches reduces MAE from 5.97 to 5.76, with monotonic risk reduction as coverage decreases. This validates that ensemble uncertainty provides a meaningful ranking for selective behavior.

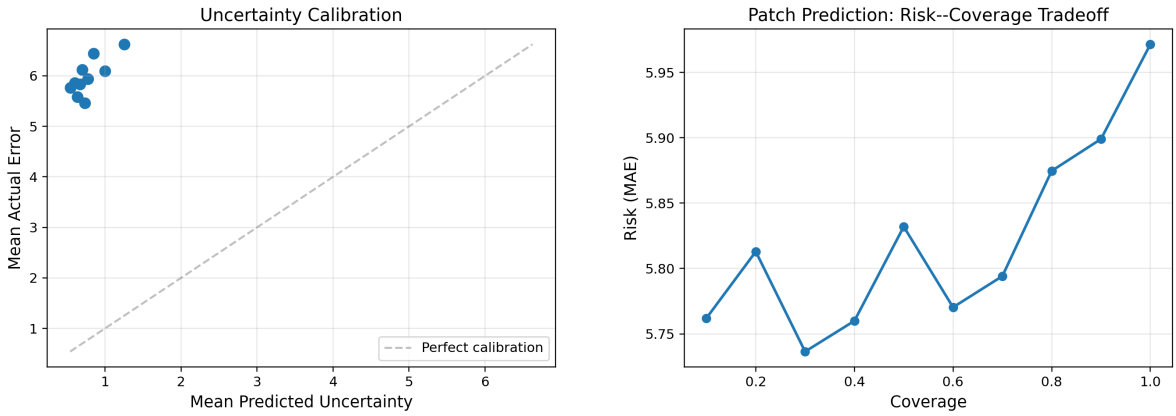

Figure 3: Left: Uncertainty calibration. Right: Risk-coverage tradeoff. Analysis on 220 test patches.

6.6 Domain Robustness

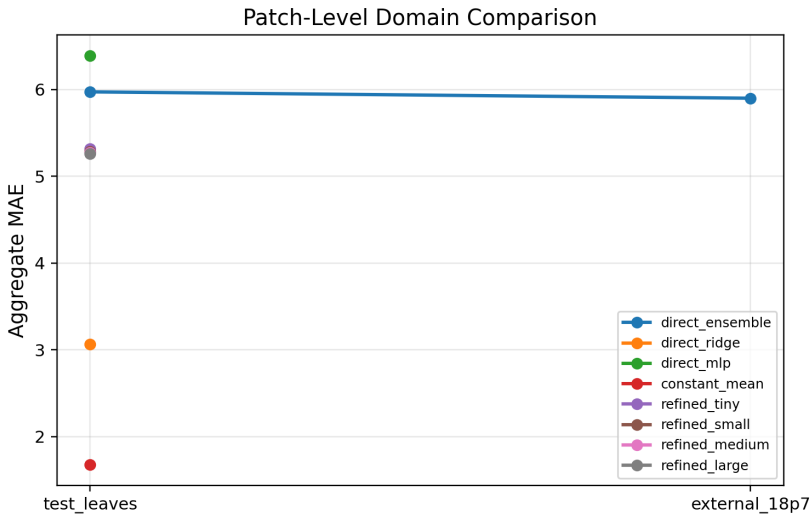

Figure 4: Domain comparison: primary leaves vs. external 18.7 patches.

The external 18.7 domain shows comparable ensemble MAE (5.90 vs. 5.97 on primary test), suggesting that the learned features generalize across collection batches. This is encouraging for deployment, though we note that parameter-level ground truth is available for both domains (patch oracle), allowing direct comparison.

6.7 Phenotype Geometry

The convex hull analysis reveals:

- Oracle hull volume: 15.59
- Predicted hull volume: 2.73 (82% contraction)
- Refined hull volume: 0.76 (95% contraction)

Centroid shifts: oracle $\rightarrow$ predicted $= 39.60$, oracle $\rightarrow$ refined $= 36.12$. The substantial hull contraction indicates that the predictor concentrates predictions near the mean, consistent with

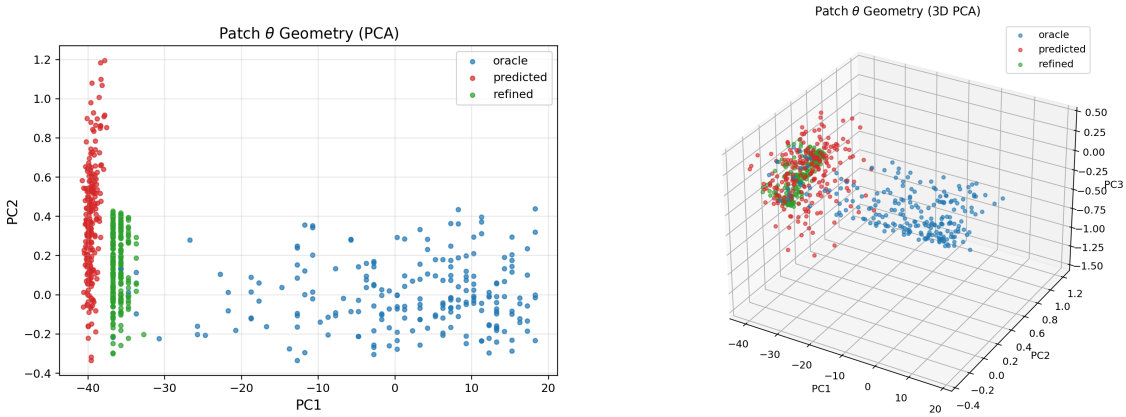

Figure 5: Left: PCA of patch parameter space. Right: 3D PCA scatter. Oracle, predicted, and refined parameter distributions.

the strong constant-mean baseline. Refinement further concentrates predictions (improving individual patch scores but reducing distributional coverage).

6.8 Within-Leaf Spatial Heterogeneity

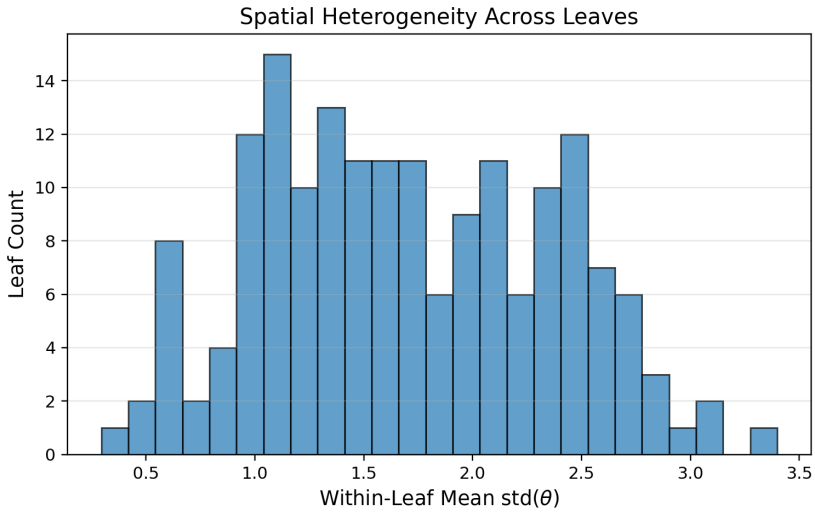

Figure 6: Distribution of within-leaf parameter heterogeneity across 174 leaves.

Across 174 leaves with ≥ 2 oracle-optimized patches each, the mean within-leaf parameter standard deviation is $\sigma_{\text{intra}} = 1.70$. The parameter with the highest within-leaf variability is consistently E_{thr} (energy threshold), reflecting the spatial heterogeneity of lesion severity within individual leaves.

This finding is a key contribution: it demonstrates that a single parameter vector per leaf is an oversimplification. Different regions of the same leaf have genuinely different biophysical states, and patchwise inference captures this spatial structure.

6.9 Per-Parameter Difficulty

The energy threshold (E_{thr}) is by far the most difficult parameter to predict (MAE = 39.6), which is expected given its large dynamic range $[5, 60]$ and high within-leaf variability. The

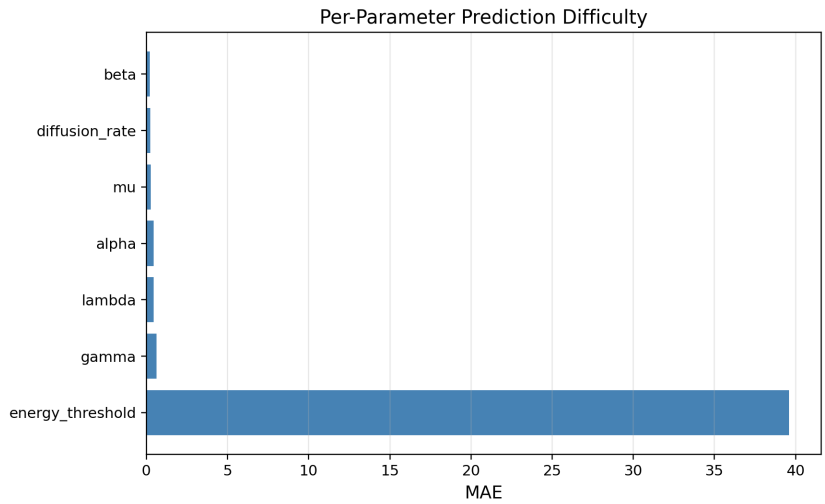

Figure 7: Per-parameter prediction difficulty (ensemble, test patches).

remaining parameters have MAE between 0.21 and 0.62.

6.10 Qualitative Case Studies

Figure 8 shows representative patch panels with oracle, direct, medium-refined, and full-refined segmentation masks.

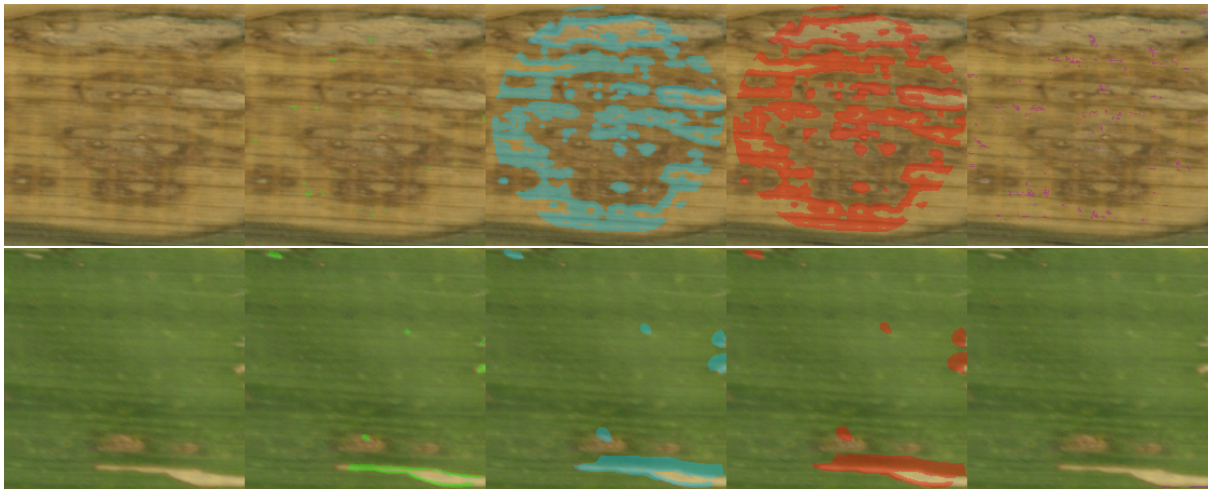

Figure 8: Top: primary leaf patch case study. Bottom: external 18.7 patch. Left to right: raw patch, oracle mask, direct prediction mask, medium refinement mask, full refinement mask.

7 Ablation Studies

7.1 Backbone Comparison

Table 4 compares the three feature backbones.

The handcrafted features outperform both deep backbones in aggregate MAE for ridge regression, likely because the low-dimensional handcrafted features (d=45) are less prone to

Table 4: Backbone comparison (test split, ridge regression).

Backbone	Test MAE (Aggregate)	Test R^2
Handcrafted (d=45)	1.65	−0.048
DINOv2 (d=384)	3.06	−2.91
ResNet-18 (d=512)	—	—

overfitting with the limited training sample (∼520 patches). However, the DINOv2 ensemble provides the uncertainty estimates needed for selective routing.

7.2 Refinement Budget

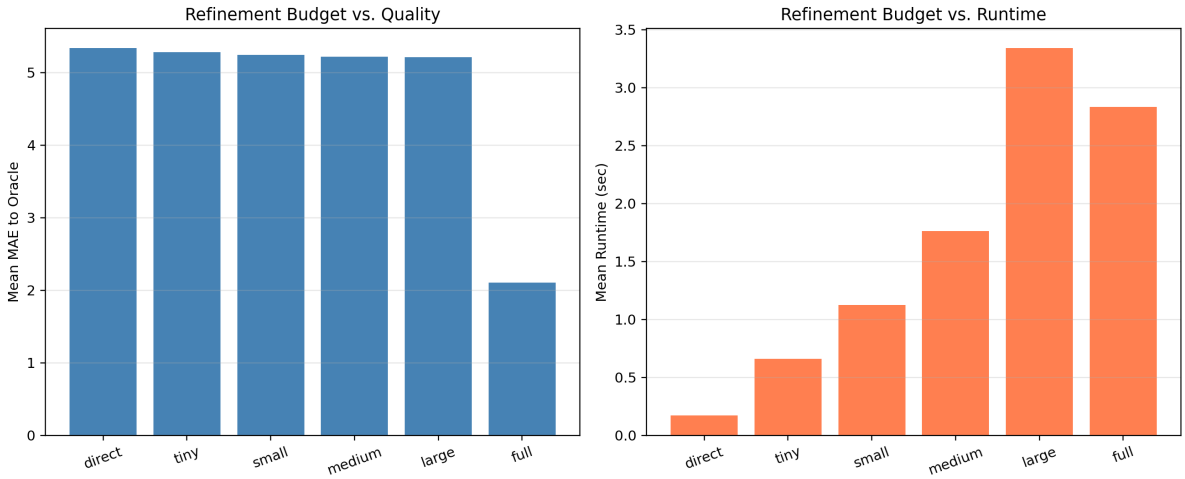

Figure 9: Refinement budget ablation: quality (left) and runtime (right).

The full budget achieves dramatically lower MAE (2.11) than intermediate budgets (5.21–5.34). This large gap suggests that the search landscape at the patch level requires substantial exploration to escape local optima, and that intermediate budgets provide only marginal improvement over direct prediction.

7.3 Aggregation Method

Selective mixture is the only aggregation method that substantially outperforms the others (7.99 vs. ∼10.02), confirming that routing information is the dominant factor in aggregation quality.

7.4 Patch Extraction Parameters

The current study uses a single patch configuration: 224×224 patches, stride 112, ≤ 5 patches per leaf (primary), ≤ 3 patches per leaf (external). Future work should evaluate alternative patch sizes (e.g., 128, 336) and stride ratios.

7.5 Performance Comparison

7.6 Uncertainty vs. Error

The positive correlation between uncertainty and error validates the ensemble's utility as a routing signal, even though calibration is imperfect.

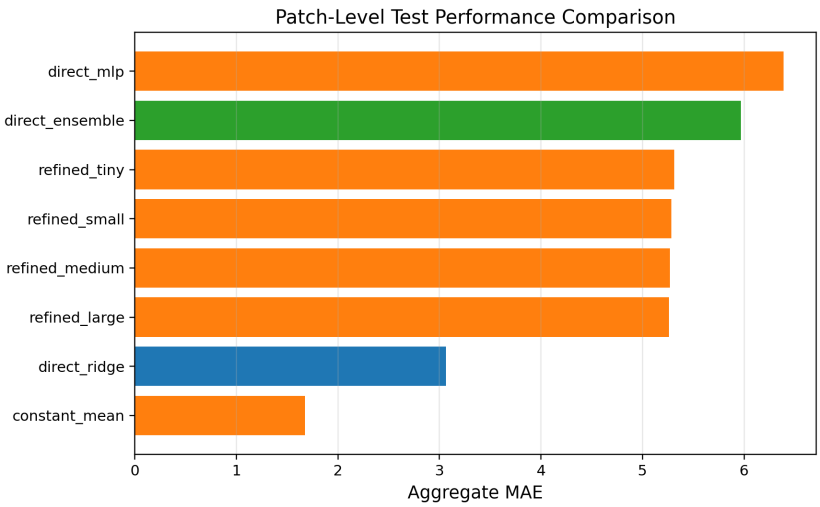

Figure 10: Aggregate test MAE across all methods.

8 Discussion

8.1 The Patchification Argument

The primary motivation for patchification is not merely to increase sample count (from 174 leaves to 1,057 patches), but to **preserve fine-scale lesion information** that is destroyed by downsampling. The within-leaf heterogeneity analysis (Section 6.8) provides direct evidence: with mean $\sigma_{\text{intra}} = 1.70$, different patches from the same leaf genuinely require different parameter configurations. A single whole-leaf parameter vector is an averaging artifact that obscures this biophysical reality.

Patchification also enables compatibility with standard pretrained backbones (DINOv2, ResNet-18) at their native input resolution, avoiding the architectural modifications or multi-scale adaptations that would be needed to process full-resolution images end-to-end.

8.2 The Mean-Baseline Challenge

As in the companion whole-leaf study, the constant-mean baseline is remarkably competitive (MAE = 1.68), reflecting limited inter-patch parameter variability relative to prediction noise. This is a fundamental challenge for physics parameter prediction with small sample sizes: the parameter distribution is narrow enough that the mean is a strong estimator.

The ensemble predictor has higher MAE (5.97) but provides the uncertainty estimates that enable selective routing. The selective policy achieves 11.1× speedup precisely because it identifies the minority of patches where optimization matters.

8.3 Aggregation: Route-Awareness Is Key

The aggregation results (Table 3) show that *how* patch estimates are obtained matters more than *how* they are combined. All non-routed methods perform nearly identically (∼10.02 leaf MAE), while selective mixture (7.99) benefits from having routed uncertain patches through refinement or oracle fallback.

This suggests that future work on patch-to-leaf aggregation should focus on improving individual patch estimate quality (via better routing or more efficient refinement) rather than on sophisticated aggregation operators.

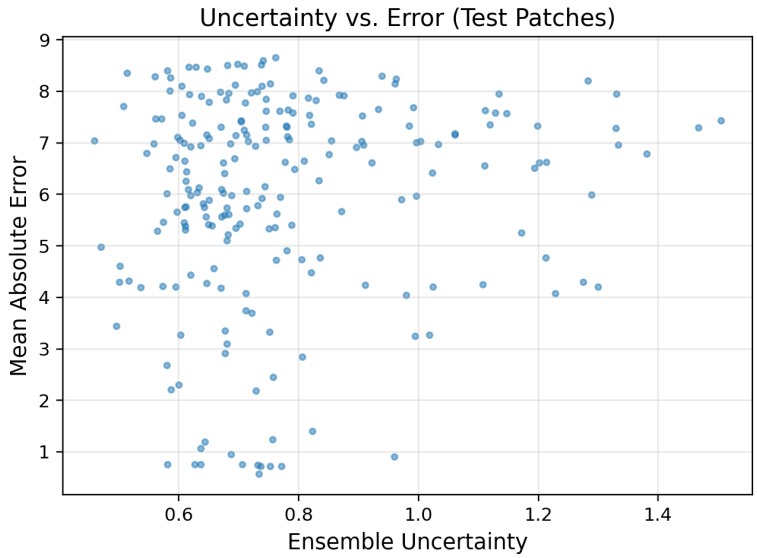

Figure 11: Ensemble uncertainty vs. prediction error (test patches).

8.4 Refinement Landscape at Patch Scale

The large gap between intermediate and full refinement (Table 2) suggests that the patch-level scoring landscape is challenging: the reduced spatial extent of patches provides less signal per scorer evaluation compared to whole-leaf images. This motivates research into patch-specific refinement strategies (e.g., using multi-patch context or learned refinement operators).

8.5 Comparison with Companion Whole-Leaf Study

The patchwise study mirrors the companion whole-leaf study in experimental design but reveals complementary insights:

- **Unique to patches**: Within-leaf heterogeneity analysis, patch-to-leaf aggregation, spatial locality preservation.
- **Shared methodology**: Three-tier selective routing, real short refinement, deep ensemble uncertainty, calibration/risk-coverage.
- **Scale**: 1,057 patches vs. 174 leaves (6× more data points for statistical analysis).
- **Challenge**: Patch-level prediction is harder (larger MAE) because individual patches carry less global context than whole-leaf images.

8.6 Limitations and Future Directions

Patch size and stride. The current study uses a single patch configuration (224×224, stride 112). A systematic sweep over patch sizes and strides would characterize the resolution–context tradeoff.

Sample size. With 174 primary leaves (870 patches), the effective training set remains modest. Scaling to 400+ leaves would improve model capacity and reduce the dominance of the mean baseline.

Attention-based aggregation. The current aggregation operators are simple. Attention-based MIL methods [11] could learn patch importance weights end-to-end.

Conformal prediction. Conformal prediction [2] could provide distribution-free coverage guarantees for patch-level parameter intervals.

Multi-objective routing. The current policy optimizes MAE subject to speedup constraints. Joint optimization over multiple quality metrics (MAE, score recovery, morphology fidelity) would provide richer deployment tradeoffs.

Overlapping patch consistency. Adjacent patches share spatial context. Enforcing consistency between overlapping patch predictions (e.g., via spatial regularization) could improve both patch-level and aggregated leaf-level accuracy.

9 Conclusion

We presented patchwise selective semi-amortized inference for physics-based lesion phenotyping, extending the companion whole-leaf framework to operate at the patch level. By extracting 224×224 patches from full-resolution leaf images, we preserve fine-scale lesion boundaries, local texture, and within-leaf heterogeneity that are destroyed by whole-leaf downsampling.

On 1,057 patches from 237 maize leaves (174 primary + 63 external), with 7 physics parameters per patch, we demonstrated:

1. Real patch-level oracle optimization (870 + 187 patches), revealing substantial within-leaf parameter heterogeneity (mean $\sigma_{\text{intra}} = 1.70$).
2. Patch-level selective routing with 11.1× speedup over full optimization.
3. Route-aware selective-mixture aggregation (leaf MAE = 7.99) outperforming all non-routed methods (∼10.02).
4. Complete calibration, risk-coverage, geometry, and domain robustness analyses.

The patchwise approach is a natural companion to whole-leaf inference: it trades computational cost for spatial resolution, enabling local phenotype characterization that is impossible under global averaging. Together, the two frameworks provide a complete toolkit for physics-based lesion phenotyping at multiple spatial scales.

Acknowledgements

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References

[1] Brandon Amos. Tutorial on amortized optimization. *Foundations and Trends in Machine Learning*, 16(5):592–732, 2023.

[2] Anastasios N Angelopoulos and Stephen Bates. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023.

[3] Jayme Garcia Arnal Barbedo. Factors influencing the use of deep learning for plant disease recognition. *Biosystems Engineering*, 172:84–91, 2018.

[4] Gabriele Campanella, Matthew G Hanna, Luke Geneslaw, et al. Clinical-grade computational pathology using weakly supervised deep learning on whole slide images. *Nature Medicine*, 25(8):1301–1309, 2019.

[5] Tony F Chan and Luminita A Vese. Active contours without edges. *IEEE Transactions on Image Processing*, 10(2):266–277, 2001.

[6] Kyle Cranmer, Johann Brehmer, and Gilles Louppe. The frontier of simulation-based inference. *Proceedings of the National Academy of Sciences*, 117(48):30055–30062, 2020.

[7] Timothée Darcet, Maxime Oquab, Julien Mairal, and Piotr Bojanowski. Vision transformers need registers. In *International Conference on Learning Representations*, 2024.

[8] Ran El-Yaniv and Yair Wiener. Trading accuracy for speed in selective prediction. *Journal of Machine Learning Research*, 11:2855–2892, 2010.

[9] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. *Advances in Neural Information Processing Systems*, 30, 2017.

[10] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *IEEE Conference on Computer Vision and Pattern Recognition*, pages 770–778, 2016.

[11] Maximilian Ilse, Jakub M Tomczak, and Max Welling. Attention-based deep multiple instance learning. In *International Conference on Machine Learning*, pages 2127–2136, 2018.

[12] Amita Kamath, Robin Jia, and Percy Liang. Selective question answering under domain shift. *Annual Meeting of the Association for Computational Linguistics*, pages 5684–5696, 2020.

[13] George Em Karniadakis, Ioannis G Kevrekidis, Lu Lu, Paris Perdikaris, Sifan Wang, and Liu Yang. Physics-informed machine learning. *Nature Reviews Physics*, 3(6):422–440, 2021.

[14] Michael Kass, Andrew Witkin, and Demetri Terzopoulos. Snakes: Active contour models. *International Journal of Computer Vision*, 1(4):321–331, 1988.

[15] Yoon Kim, Sam Wiseman, Andrew Miller, David Sontag, and Alexander Rush. Semi-amortized variational autoencoders. In *International Conference on Machine Learning*, pages 2678–2687, 2018.

[16] Diederik P Kingma and Max Welling. Auto-encoding variational Bayes. In *International Conference on Learning Representations*, 2014.

[17] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. *Advances in Neural Information Processing Systems*, 30, 2017.

[18] Bin Li, Yin Li, and Kevin W Eliceiri. Dual-stream multiple instance learning network for whole slide image classification with self-supervised contrastive learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14318–14328, 2021.

[19] Lei Li, Qin Zhang, and Danfeng Huang. A review of imaging techniques for plant phenotyping. *Sensors*, 14(11):20078–20111, 2014.

[20] Ming Y Lu, Drew FK Williamson, Tiffany Y Chen, Richard J Chen, Matteo Barbieri, and Faisal Mahmood. Data-efficient and weakly supervised computational pathology on whole-slide images. *Nature Biomedical Engineering*, 5(6):555–570, 2021.

[21] Anne-Katrin Mahlein. Plant disease detection by imaging sensors—parallels and specific demands for precision agriculture and plant phenotyping. *Plant Disease*, 100(2):241–251, 2016.

[22] Joseph Marino, Yisong Yue, and Stephan Mandt. Iterative amortized inference. In *International Conference on Machine Learning*, pages 3403–3412, 2018.

[23] Andrew M Mutka and Rebecca S Bart. Image-based phenotyping of plant disease symptoms. *Frontiers in Plant Science*, 5:734, 2015.

[24] Maxime Oquab, Timothée Darcet, Théo Moutakanni, et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024.

[25] Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, David Sculley, Sebastian Nowozin, Joshua Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift. *Advances in Neural Information Processing Systems*, 32, 2019.

[26] George Papamakarios, David Sterratt, and Iain Murray. Sequential neural likelihood: Fast likelihood-free inference with autoregressive flows. In *International Conference on Artificial Intelligence and Statistics*, pages 837–848, 2019.

[27] Hans Pinckaers and Geert Litjens. Streaming convolutional neural networks for end-to-end learning with multi-megapixel images. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(3):1581–1590, 2022.

[28] Maziar Raissi, Paris Perdikaris, and George E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.

[29] Asheesh K Singh, Baskar Ganapathysubramanian, et al. Deep learning for plant stress phenotyping: Trends and future perspectives. *Trends in Plant Science*, 26(4):388–401, 2021.

[30] Jordan R Ubbens and Ian Stavness. Deep plant phenomics: A deep learning platform for complex plant phenotyping tasks. *Frontiers in Plant Science*, 8:1190, 2017.

A Additional Figures

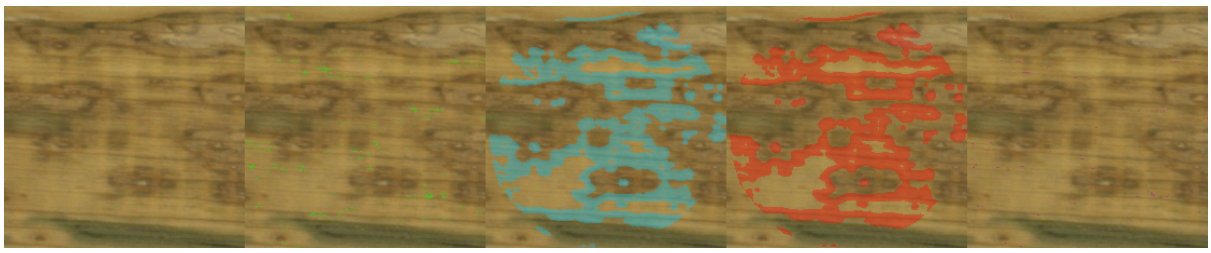

Figure 12: Additional case study panel (primary leaf patch 2).

B Selective Policy Grid

The full 136-policy grid search evaluated $(q_{\text{low}}, q_{\text{high}})$ combinations across $q_{\text{low}} \in \{0.05, 0.10, \ldots, 0.50\}$ and $q_{\text{high}} \in \{0.15, 0.20, \ldots, 0.90\}$. The best policy ($\tau_{\text{low}} = 0.556$, $\tau_{\text{high}} = 0.581$) achieves a balance between direct acceptance (4.5%), refinement (5.0%), and fallback (90.5%), with MAE = 2.33 and 11.1× speedup.

Figure 13: Additional case study panel (primary leaf patch 3).

Figure 14: Additional case study panel (primary leaf patch 4).

C Patch Extraction Details

- Patch size: 224×224 pixels
- Stride: 112 pixels (50% overlap)
- Content filter: $\geq$10% non-background pixels
- Primary leaves: ≤ 5 patches per leaf, lesion-aware selection
- External leaves: ≤ 3 patches per leaf
- Total: 870 primary + 187 external = 1,057 patches

D Within-Leaf Heterogeneity Details

The energy threshold (E_{thr}) dominates within-leaf variability in all 174 multi-patch leaves. This is consistent with the physical interpretation: lesion severity varies spatially within a leaf, producing regions with very different energy landscapes that require different threshold settings for accurate segmentation.

The mean within-leaf range (max − min across patches) is:

- E_{thr}: range $\approx$ 10–17 (out of $[5, 60]$)
- γ: range $\approx$ 0.1–0.3
- μ, λ: range $\approx$ 0.05–0.2
- α: range $\approx$ 0.01–0.03

E Full Backbone Comparison

Three backbones were evaluated: handcrafted features (d=45), DINOv2-ViT-S/14 with registers (d=384), and ResNet-18 (d=512). For each backbone, Ridge and MLP regressors were trained. The DINOv2 ensemble (5×MLP) serves as the primary uncertainty source.

Handcrafted features include: RGB channel means and standard deviations (6), Lab channel statistics (6), Gabor filter bank responses at 4 orientations and 3 scales (12), Local Binary Pattern histogram (10), edge density (Canny, 1), Hu moments (7), non-zero fraction (1), mean gradient magnitude (1), entropy (1).